Career Development Center

Lab Instructions for Programming Practice

1. General Guidelines

- Each student must write, compile, and run every program in the lab list (90 programs).
- Programs should be written in C language (unless told otherwise).
- Each program must include:
  - Proper comments (problem statement, logic explanation, author name, date).
  - Input/output samples in comments.
  - Clean indentation and formatting.

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2. Record Maintenance

- Students should maintain two forms of record:
  1. GitHub Repository (for personal portfolio).
  2. Online Coding Platform Account (for solving algorithmic equivalents).

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3. GitHub Instructions

1. Create a new repository named:
2. C-Programming-Lab

or

Programming-90-Questions

3. Inside the repo, create folders Lab1, Lab2, …, Lab10.
4. Save each program with a meaningful filename, for example:
   - Lab1_Prog1_MarksPercentage.c
   - Lab4_Prog32_Factorial.c
5. Add a README.md file that contains:
   - Student details (Name, Roll No, Course).
   - List of all 90 problems.
   - Status (✓ Completed / ⏳ Pending).
6. After writing each program:
7. git add filename.c
8. git commit -m "Added Lab1 Program1 - Marks & Percentage"
9. git push origin main

10. By the end of semester, GitHub repo will serve as a digital lab record + portfolio for placements.

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4. Online Coding Platform Practice

- Every student must also practice equivalent problems on online platforms:
  - LeetCode
  - GeeksforGeeks Practice
  - **HackerRank**

Mapping Strategy:

- Lab 1–4 (Basics, Loops, Conditions): Solve beginner problems on HackerRank "C" track.
- Lab 5–6 (Recursion & Patterns): Solve recursion/pattern questions on GeeksforGeeks.
- Lab 7 (Arrays): Solve Easy/Medium array questions on LeetCode (Two Sum, Rotate Array, etc.).
- Lab 9 (Matrices): Practice matrix problems on GeeksforGeeks (Spiral Traversal, Rotate Matrix, etc.).
- Lab 10 (Strings & Structures): Solve string-based problems on LeetCode (Valid Palindrome, Anagram, etc.).

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5. Submission & Evaluation

- **Weekly Check:**
  - Lab Instructor will check GitHub commits (each student must commit regularly, not all at once).
  - At least 5 programs must be pushed every week.
- **Monthly Review:**
  - Students must show progress on GeeksforGeeks/LeetCode profile (screenshot or live demo).
- **Final Evaluation:**
  - GitHub repo must contain all 90 programs.
  - At least 20 problems solved online (screenshot proof).

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6. Additional Best Practices

- Always use meaningful variable names (e.g., marks, percentage) not a, b.
- Use functions wherever possible (modular programming).
- Write test cases for your programs with multiple inputs.
- Collaborate with classmates by reviewing each other's GitHub repos.

List of Programs for B. Tech First Year Lab

Lab 1: Basics (Input/Output, Operators, Conversion)

1. Write a program to accept marks of 5 subjects, find the sum and calculate percentage.
2. Write a program to calculate Simple Interest and Compound Interest.
3. Write a program to calculate the area and circumference of a circle for a given radius.
4. Write a program to convert temperature between Celsius and Fahrenheit.
5. Write a program to convert distance from kilometers to miles and miles to kilometers.
6. Write a program to convert days into years, months and remaining days (assume 365 days = 1 year, 30 days = 1 month).
7. Write a program to calculate HRA, DA, and Gross Salary for a given basic salary.
8. Write a program to find the perimeter of a rectangle and the area of a square.
9. Write a program to perform all arithmetic operations (+, −, ×, ÷, %) on two numbers.
10. Write a program to check whether a given character is a vowel or consonant.

Lab 2: Conditions (if-else, relational operators)

11. Write a program to swap values of two variables using a third variable.
12. Write a program to check whether two numbers are equal or not.
13. Write a program to find the greatest among three numbers.
14. Write a program to check whether a given number is even or odd.
15. Write a program to find the roots of a quadratic equation.
16. Write a program to check whether a given character is an alphabet, digit, or special symbol.
17. Write a program to check whether a number is positive, negative or zero.
18. Write a program to check whether three sides can form a valid triangle and display its type.
19. Write a program to calculate electricity bill based on units consumed (using slab rates).
20. Write a program to create a simple calculator using if-else ladder.

Lab 3: Decisions & Switch

21. Write a program to check whether a given year is a leap year or not.
22. Write a program to accept marks of 5 subjects, calculate percentage and display grade (A/B/C/D).
23. Write a program to create a calculator using switch-case.

24. Write a program to display the weekday name for a given day number (1–7).
25. Write a program to display the number of days in a given month (1–12).
26. Write a program to check whether a character is vowel or consonant using switch.
27. Write a program to check eligibility for voting (age ≥ 18).
28. Write a program to categorize a person based on age (Child/Teen/Adult/Senior).
29. Write a program to create a menu-driven program to calculate area of geometric shapes.
30. Write a program to create a menu-driven program for temperature conversion (C↔F).

Lab 4: Loops (for, while, do-while)

31. Write a program to find the sum of natural numbers up to N.
32. Write a program to find the factorial of a given number.
33. Write a program to calculate the sum of even and odd numbers up to N.
34. Write a program to print the multiplication table of a given number.
35. Write a program to print the first N natural numbers.
36. Write a program to reverse the digits of a number using loop.
37. Write a program to count the number of digits in a number.
38. Write a program to check whether a number is a palindrome or not.
39. Write a program to print all factors of a given number.
40. Write a program to calculate power (x^y) without using the pow() function.

Lab 5: Series & Recursion

41. Write a program to generate Fibonacci series up to N terms (loop).
42. Write a program to check whether a number is prime or not.
43. Write a program to find the sum of digits of a number.
44. Write a program to check whether a number is an Armstrong number.
45. Write a program to check whether a number is a Strong number.
46. Write a program to check whether a number is a Perfect number.
47. Write a program to print all prime numbers in the range 1–N.
48. Write a program to generate Fibonacci series using recursion.
49. Write a program to find factorial of a number using recursion.
50. Write a program to find GCD and LCM of two numbers.

Lab 6: Patterns

51. Write a program to reverse a number using recursion.

52. Write a program to print all Armstrong numbers between 1 and 100.
53. Write a program to convert a binary number to decimal and decimal to binary.
54. Write a program to print Pascal's Triangle.
55. Write a program to print Floyd's Triangle.
56. Write a program to print a pyramid of numbers.
57. Write a program to print a pyramid of stars.
58. Write a program to print a diamond pattern.
59. Write a program to print a hollow square pattern.
60. Write a program to print an inverted half pyramid.

Lab 7: Arrays (1D, Searching, Sorting)

61. Write a program to accept elements in an array and find their sum.
62. Write a program to add two arrays and store result in a third array.
63. Write a program to find the minimum and maximum element of an array.
64. Write a program to search an element in an array using Linear Search.
65. Write a program to sort an array using Insertion Sort.
66. Write a program to sort an array using Selection Sort.
67. Write a program to sort an array using Bubble Sort.
68. Write a program to search an element in an array using Binary Search.
69. Write a program to print the second largest and second smallest element in an array.
70. Write a program to reverse an array in-place.

Lab 8 : Matrices & 2D arrays

71. Write a program to add two matrices.
72. Write a program to multiply two matrices.
73. Write a program to find the sum of diagonal elements of a matrix.
74. Write a program to find the transpose of a matrix.
75. Write a program to print the upper and lower triangular parts of a matrix.
76. Write a program to check whether a matrix is symmetric or not.
77. Write a program to check whether a matrix is sparse or not.
78. Write a program to print the boundary elements of a matrix.
79. Write a program to find the row-wise and column-wise sum of a matrix.
80. Write a program to find the maximum element in each row and each column of a matrix.

Lab 9: Strings, Pointers, Files, Structures

81. Write a program to implement strlen() function.
82. Write a program to implement strcpy() function.
83. Write a program to implement strcat() function.
84. Write a program to count vowels and consonants in a string.
85. Write a program to check whether a string is palindrome or not.
86. Write a program to reverse a string without using library functions.
87. Write a program to count frequency of each character in a string.
88. Write a program to remove all spaces from a string.
89. Write a program to convert a string to uppercase and lowercase.
90. Write a program to search for a substring inside a string (manual strstr).